

GitHub Repository Link: <https://github.com/IgnacioLY9/CS5833Project/>

GitHub PR Link: <https://github.com/Blobscan/blobscan/pull/982>

YouTube Demo: <https://youtu.be/9WzsHFktUIU>

Group Members and Contributions:

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As this was an individual project, there was no collaboration or communication with other group members.

# Project Report

## Introduction

The purpose of this project was to contribute to an existing Ethereum or Web3 project/application. Blobscan is an open-source blockchain explorer that primarily focuses on Ethereum blobs. Blobscan's public API that allows users to fetch data from Blobscan's database is not very developed and lacks good documentation. This project improved the Blobscan repository by adding new API endpoints and improving the documentation for certain existing endpoints. Additionally, the installation guide for local use was updated.

## Background

A blob is an Ethereum data type that allows for short term storage of data. Blobs were introduced into the Ethereum blockchain in order to increase scalability. Their data and transactions cannot be accessed by the Ethereum Virtual Machine, but they are still included in the consensus. Because they are meant for short term storage, their data can safely be deleted after a short period of time [1]. Because users now have better ways of publishing the data associated with a rollup, rather than permanently storing it in the L1 chain, users can pay less in gas fees.

Because of the utility of blobs for the Ethereum blockchain, the blockchain explorer Blobscan is important. Blobscan, unlike other blockchain explorers, focuses primarily on blobs. Using Blobscan, Ethereum users and developers can easily look at information about blobs, their associated gas prices, and the transactions they contain. Additionally, Blobscan provides users with a public API that can be used to directly query information from the Blobscan database

about Ethereum blobs. However, Blobscan's API is very rudimentary. This project aims to contribute to the Blobscan API.

## Methodology and Contributions

On Blobscan's GitHub repository (<https://github.com/Blobscan/blobscan>), some users expressed disappointment because of Blobscan API's limitations. Some of the complaints regarding how endpoints functioned were entirely due to a lack of documentation on the API page. The endpoints behaved exactly as designed, but the users didn't know that. In order to contribute to this repository, Quicknode was used to connect to the Sepolia test network. Endpoints were created following the OpenAPI framework which allows the Blobscan API page built using Express and Swagger to automatically build documentation of the endpoint and allow users to use the endpoint. The following methods and their test files were added to the repository in order to improve the API:

### Gas Price Summary

Originally, users were only able to query information about average blob gas prices. A user would not be able to directly query additional information about the distribution of gas prices. The Blobscan repository keeps track of some overall statistics about blobs. These statistics are updated every 15 minutes and stored in the database. Whenever a user wants to see the overall statistics, the API fetches the results from the table in the database. This new API endpoint allows users to retrieve a 5 number summary about the distribution of blob gas prices. This was implemented by adding a new table in the database to store the summary. Just like the original table, it is updated every 15 minutes, and the entries are returned to the user whenever a request is made.

### Blob Transactions

This endpoint returns the transactions contained within a blob. It searches for a specific blob by versioned hash or kzg commit. This allows users to quickly fetch information about the transactions contained within a blob without having to arrive at this information indirectly through other API endpoints.

### Block Gas Price

This endpoint returns the total blob gas price of a specific block. This endpoint does have one important limitation. The Blobscan database only stores blocks that contain blobs. So, a valid block ID could be entered, but the endpoint will return a 404 error. Similarly, the existing endpoint that searches for a block based on its ID has the same limitation. This was not mentioned anywhere in the original documentation. This limitation is now mentioned clearly in both the documentation of the new and original endpoints.

## Gas Time Data

This endpoint retrieves the average blob gas price of certain time ranges. Specifically, a user can get information about the all-time average gas price, or the average gas price of the most recent day, week, month, etc. This allows users to see how the average blob gas price has changed over a time range. This endpoint also returns the timestamps associated with the values returned. This way, the user can see the specific time the average gas price was calculated at.

## Installation Guide Improvements

Initial work on this project was slow due to issues trying to run the repository. Blobscan publishes Docker containers, but these could not be rebuilt whenever changes were made to the repository. As a result, the repository had to be run locally. Following the installation guide on the Blobscan documentation page didn't work. After a lot of troubleshooting, the code could be run locally. The original installation guide appears to be out of date, so it has been updated.

## Results

Original blobs API endpoints:

| blobs |                           |                                                                    |
|-------|---------------------------|--------------------------------------------------------------------|
| POST  | /blobs/weavevm-references | Creates WeaveVM references for a given set of blobs.               |
| GET   | /blobs                    | retrieves all blobs.                                               |
| GET   | /blobs/{id}               | retrieves blob details for given versioned hash or kzg commitment. |
| GET   | /blobs/{id}/data          | Retrieves blob data for given blob id.                             |

Updated blobs API endpoints:

| blobs |                           |                                                                         |
|-------|---------------------------|-------------------------------------------------------------------------|
| POST  | /blobs/weavevm-references | Creates WeaveVM references for a given set of blobs.                    |
| GET   | /blobs                    | retrieves all blobs.                                                    |
| GET   | /blobs/{id}               | retrieves blob details for given versioned hash or kzg commitment.      |
| GET   | /blobs/transactions/{id}  | retrieves blob transactions for given versioned hash or kzg commitment. |
| GET   | /blobs/{id}/data          | Retrieves blob data for given blob id.                                  |

Original blocks API endpoints:

| blocks |              |                                                         |
|--------|--------------|---------------------------------------------------------|
| GET    | /blocks      | retrieves all blocks.                                   |
| GET    | /blocks/{id} | retrieves block details for given block number or hash. |

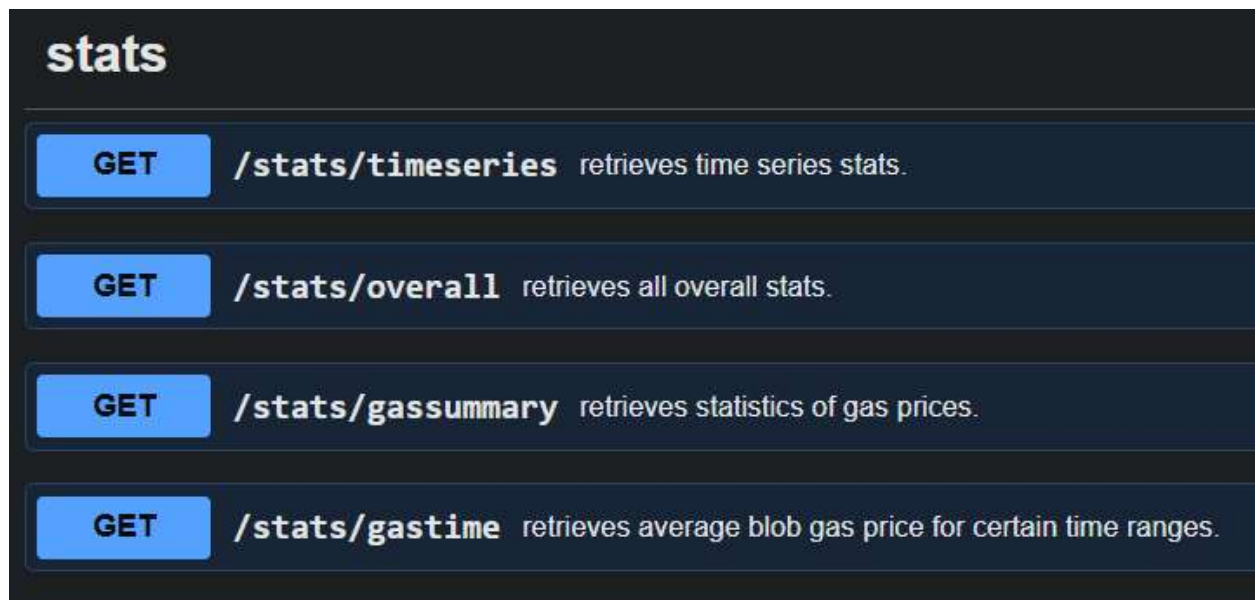
Updated blocks API endpoints:

| blocks |                       |                                                                     |
|--------|-----------------------|---------------------------------------------------------------------|
| GET    | /blocks               | retrieves all blocks.                                               |
| GET    | /blocks/gasPrice/{id} | retrieves blob gas price of a block for given block number or hash. |
| GET    | /blocks/{id}          | retrieves block details for given block number or hash.             |

Original stats API endpoints:



Updated stats API endpoints:



The code for the added API endpoints can be found within `/packages/api/src/routers`. The specific files added were:

- `/stats/getGasSummary.ts`
- `/stats/getGasTime.ts`
- `/blob/getBlobTransactions.ts`
- `/block/getBlockGasPrice.ts`

The corresponding test file for each added API endpoint can be found within `/packages/api/test`. The specific files added and their associated successful tests were:

- `/stats/getGasSummary.test.ts`

```
✓ |api| test/stats/getGasSummary.test.ts (1 test) 387ms
```

- `/stats/getGasTime.test.ts`

```
✓ |api| test/stats/getGasTime.test.ts (12 tests) 747ms
```

- /blob/getBlobTransactions..test.ts

```
✓ |api| test/blob/getBlobTransactions.test.ts (3 tests) 225ms
```

- /block/getBlockGasPrice.test.ts

```
✓ |api| test/block/getBlockGasPrice.test.ts (12 tests) 458ms
```

The required updates to the database schema can be found in `/packages/db/prisma/schema.prisma`. The model `GasStats` previously did not exist and was added.

A PR was submitted to the original Blobscan repository. Only the updated documentation and installation guide were included. The API endpoints and their tests were not included because of the required changes to the database. To view the API endpoints functioning, please watch the demonstration video linked at the top of this report.

## Conclusion

This project contributed to the open-source Ethereum blockchain explorer Blobscan. Blobscan allows users and developers to analyze Ethereum blobs through their website or using their public API. This project added four new API endpoints along with corresponding tests and documentation. Final evaluation differs slightly from project proposal, but a PR was still submitted to the Blobscan GitHub repository.

## References

1. Vitalik Buterin, Dankrad Feist, Diederik Loerakker, George Kadianakis, Matt Garnett, Mofi Taiwo, Ansgar Dietrichs, "EIP-4844: Shard Blob Transactions," *Ethereum Improvement Proposals*, no. 4844, February 2022. Available: <https://eips.ethereum.org/EIPS/eip-4844>.